

Unit 11, Lesson 4: Informally Fitting a Line to Bivariate Data

Benchmarks

MA.8.DP.1.2 Given a scatter plot within a real-world context, describe patterns of association.

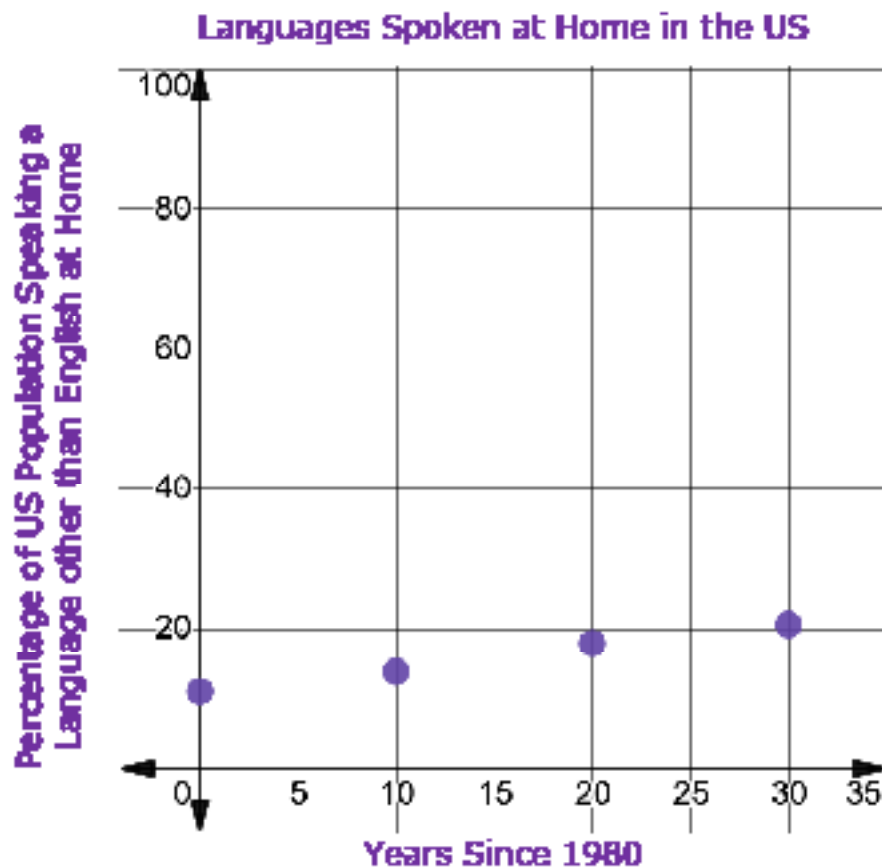
MA.8.DP.1.3 Given a scatter plot with a linear association, informally fit a straight line.

Learning Targets

- ☐ I can describe patterns of associations in data on a scatter plot.
- ☐ I can informally fit a straight line to data on a scatter plot.

11.4.1 Warm-Up: Notice and Wonder

The graph shows the percentage of the US population 5 years of age and older who speak a language other than English at home, according to the U.S. Census, since 1980.

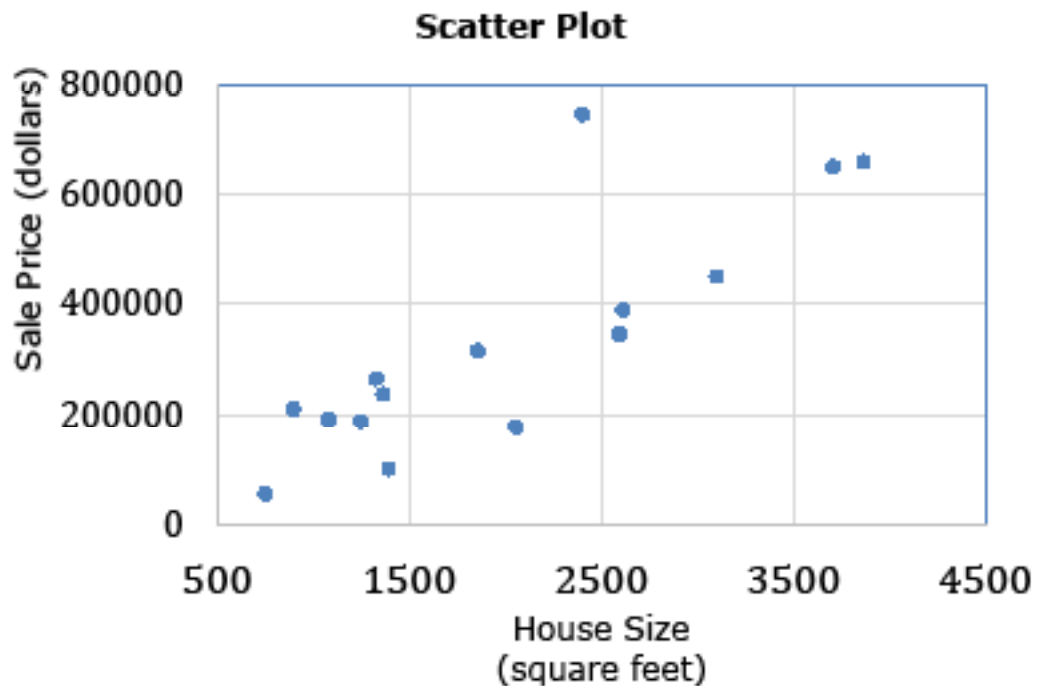


1. What do you notice?
2. What do you wonder?

11.4.2 Exploration: Lines of Fit

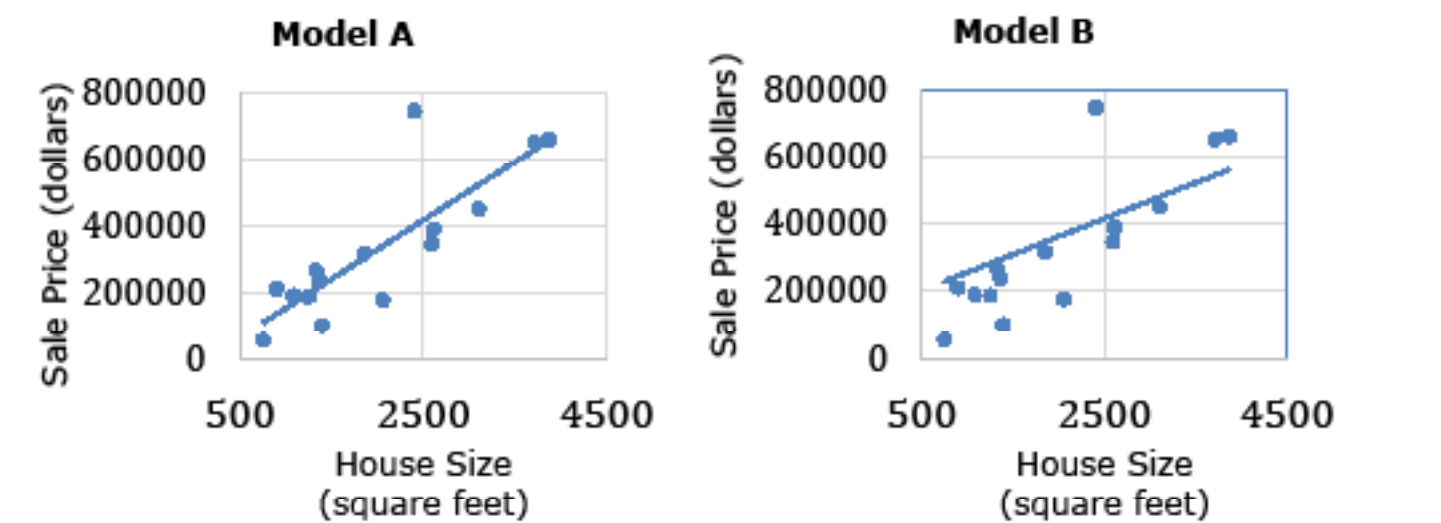
Work with your partner to complete the following.

1. The scatter plot shows the house size and price for the 15 houses sold in Orlando, FL, in 2020.



- a. One of the data points on the scatter plot is (2056, 176900). Interpret this point in context.

Two different lines, or models, are used to represent the relationship between house size and price on the scatter plot. Both models are shown.



- b. What do you notice about the models?
- c. Work with your partner to describe the similarities and differences between the models.

Similarities	Differences

This type of model is called a ***line of fit***.



The ***line of fit*** is a line drawn on a scatter plot to estimate the relationship between two sets of data. It is also known as a trend line.

- d. Discuss with your partner which line appears to better estimate the relationship between house size and sale price. Summarize your discussion.

11.4.3 Bring It Together: Fitting Lines to Model Bivariate Data

1. Complete the statements.

Most of the data points were below the line in Model

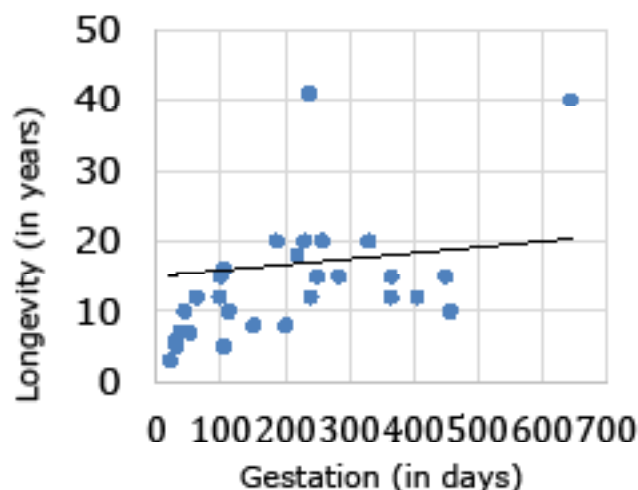
- ☐ A.
☐ B.

Model ☐ A
☐ B had an approximately equal number of data points above and below the

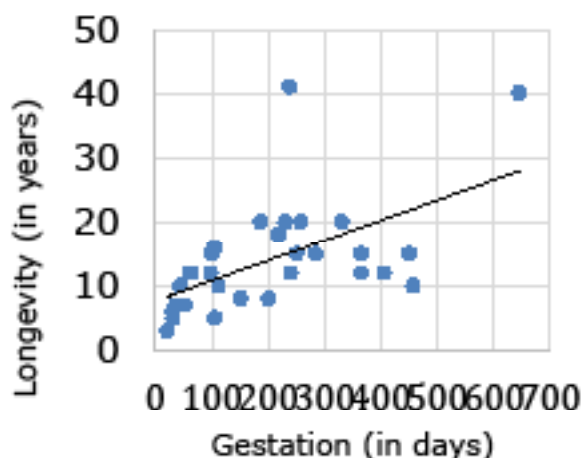
line. Therefore, ☐ A
☐ B better models the relationship between house size and sale price.

2. The two scatter plots show the same set of data summarizing the average gestation period, in days, and average longevity, in years, for a sample of 30 animals as reported in *The World Almanac and Book of Facts* (McGeveran, 2006). A gestation period is the amount of time for which an animal is pregnant. Longevity refers to how long an animal lives. Each scatter plot shows a different line to model the relationship between gestation period and longevity.

Line of Fit A



Line of Fit B



- a. Explain which of the lines is a better fit to the data.

b. Complete the statements.

There is a ☐ positive,
☐ negative,

☐ linear
☐ non-linear

association between the gestation period and longevity of an animal. The association appears to be

☐ weak.
☐ moderate.
☐ strong.

c. Discuss with your partner how the slope of the line and fit to the data relates to the patterns of association in a scatter plot that appears linear. Then, summarize your discussion.

Sometimes data includes **outliers**.



An **outlier** is a value that is much higher or much lower than the other values in a set of data.

d. Circle the point(s) on the scatter plot that appear to be outliers in the data set.

e. Complete the statement.

Most of the animals included in the data set follow a similar trend, modeled

by the

- ☐ Line of Fit A.
- ☐ Line of Fit B.

However, there

- ☐ is 1
- ☐ are 2
- ☐ are 3

animal(s) included in the

data set who appear to

- ☐ live
- ☐ be pregnant for

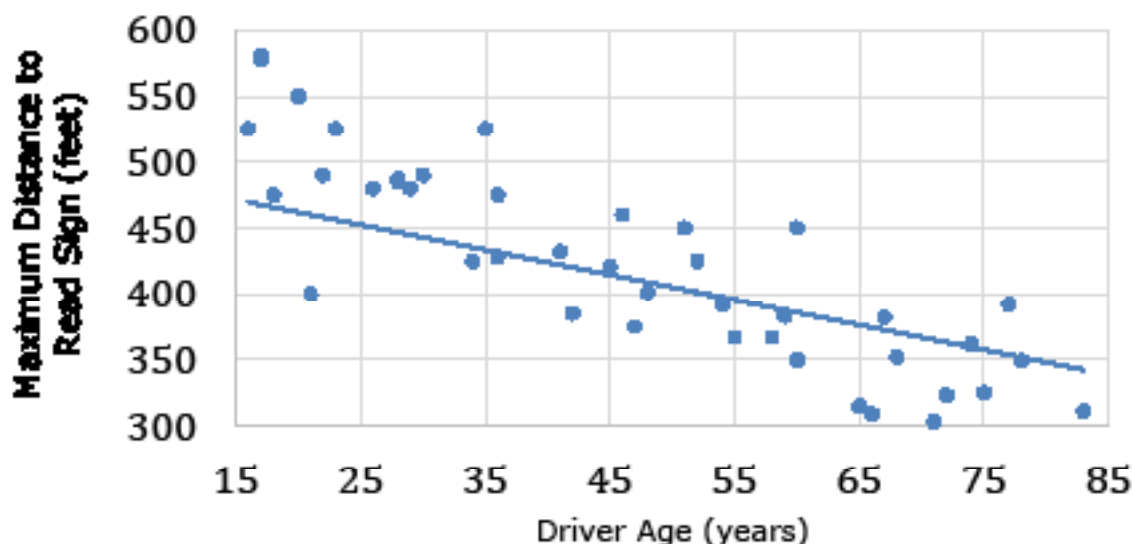
much longer

than the others, regardless of

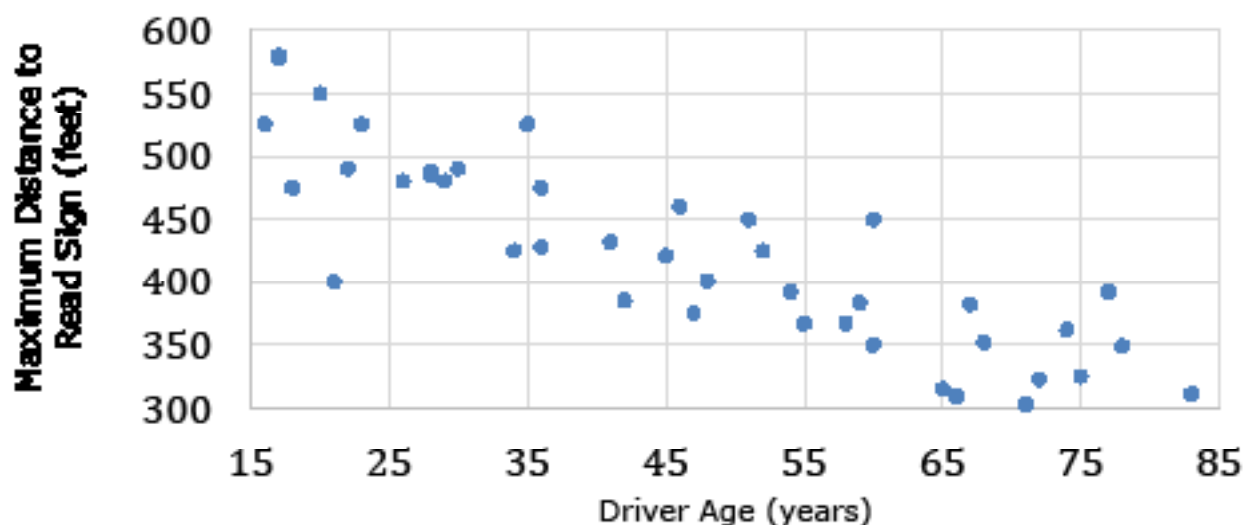
- ☐ how long they live.
- ☐ the length of the gestation period.

11.4.4 Guided Practice: Informally Fitting a Line

1. The scatter plot shows the relationship between driver age and the maximum distance to read a sign.



- a. Explain if the line shown is a good fit for the data.
- b. Using a straightedge, draw a line that better fits the data.



c. Complete the statement.

As the independent variable

- ☐ increases,
- ☐ decreases,

the dependent variable

tends to

- ☐ increase.
- ☐ decrease.

Therefore, there is a

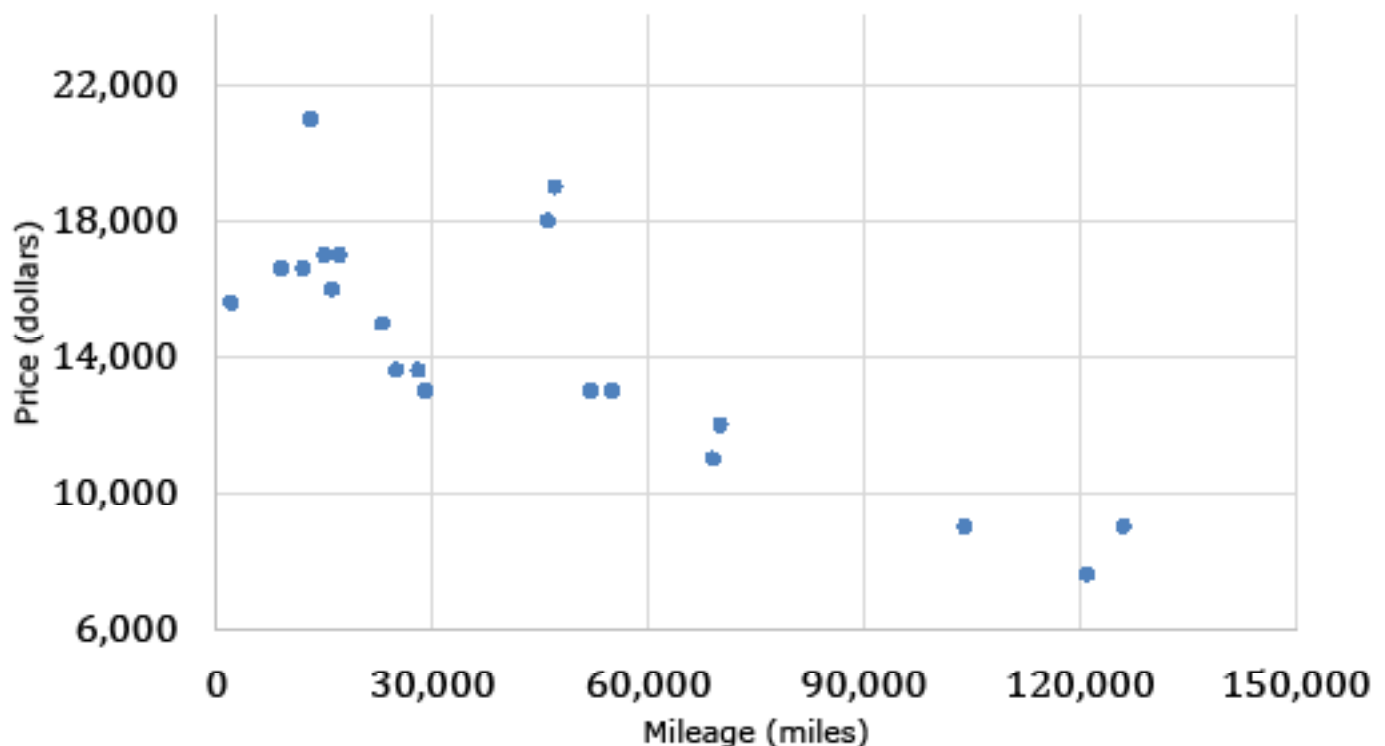
- ☐ positive
- ☐ negative

pattern of association between the

variables, so the slope of the line fit to the data is

- ☐ positive.
- ☐ negative.

2. Wyatt's Auto World sells used cars. The scatter plot displays the data for 20 recently sold used cars by comparing the mileage of the car and its price.



a. Using a straightedge, draw a line of fit for the scatter plot.

b. Complete the statement.

The slope of the line of fit is ☐ positive,
☐ negative, meaning as the mileage on a
car ☐ increases,
☐ decreases, the price Wyatt's Auto World can sell it
for ☐ increases.
☐ decreases.

c. Describe any patterns of association. Use at least two words from the word bank in your description.

decrease(s)

increase(s)

linear

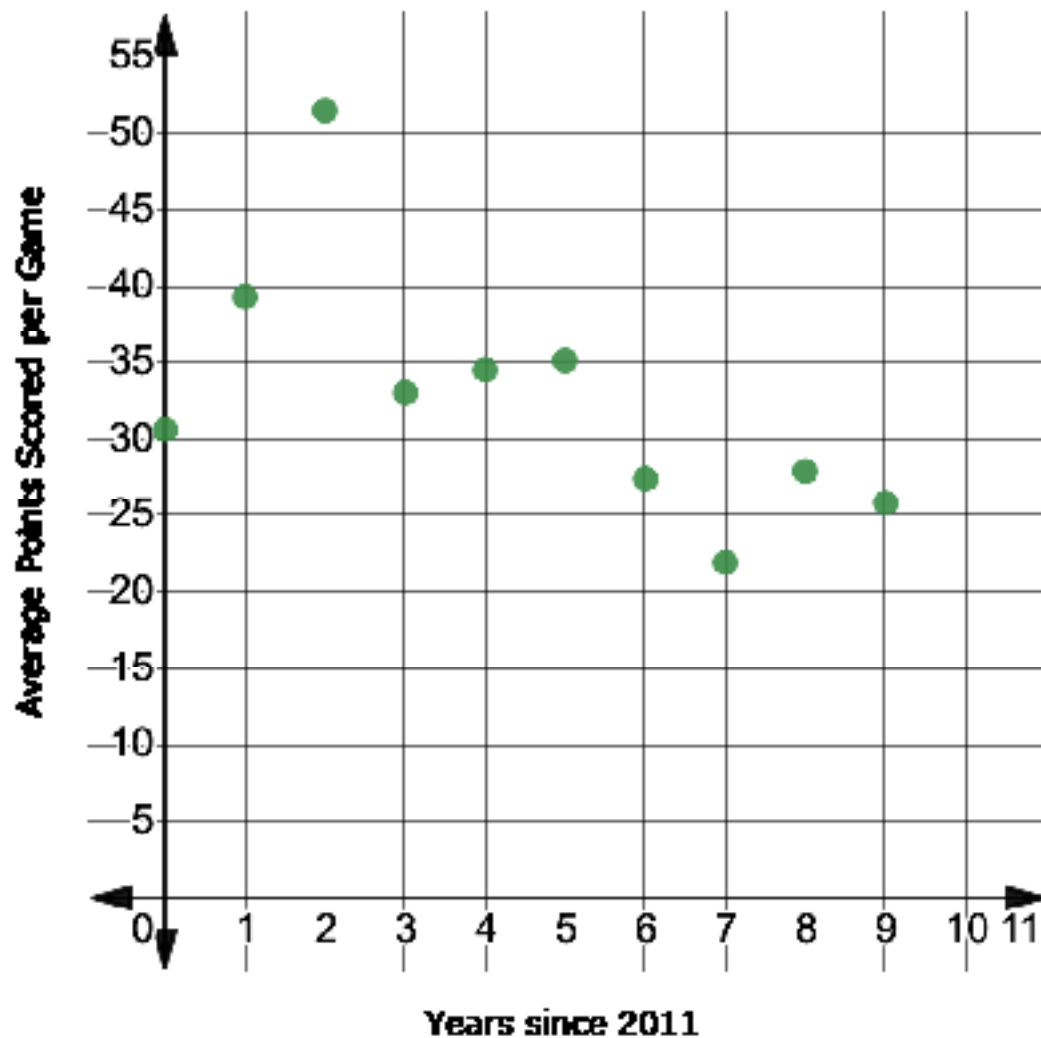
non-linear

positive

negative

11.4.5 Your Turn!

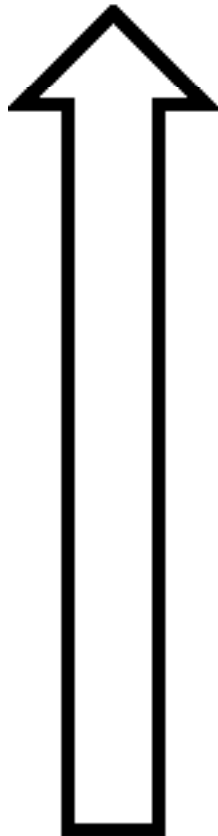
1. The scatter plot shows the average points per game scored by the Florida State University (FSU) Seminoles from 2011–2020.



- a. Use a straightedge to draw a line that models the relationship between the variables.
- b. Describe the patterns of association. Include if any points appear to be outliers in your description.

11.4.6 Wrap-Up: Reflection

1. **Color in the arrow up to the statement that best describes your current understanding.**



I'm so confident, I could explain this to someone else!

I can get to the right answer, but I don't understand well enough to explain it yet.

I understand some of this, but I don't understand all of it yet.

I tried hard and listened, but I am finding this challenging. I will make sure that I get help with this next lesson.

I do not understand any of this yet. There are things I could do to be a better learner next lesson.

